SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Deep Learning COURSE CODE: MLA0404

NAME: P. GAYATHRI REG.NO:192425277

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

I __P.GAYATHRI/192425277____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Answer:

Error Analysis Task

Question:

A neural network predicts student scores with an **MSE of 0.85 during training** and **5.20 during testing**. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer

The large difference between the training Mean Squared Error (MSE = 0.85) and testing Mean Squared Error (MSE = 5.20) indicates that the neural network is **overfitting** the training data. The model performs very well on the training set but fails to generalize to unseen test data.

Possible Reasons

1. **Overfitting**
    - The model has learned the training data, including noise and unnecessary patterns.
    - As a result, it performs poorly on new data.
2. **Complex Neural Network**
    - Too many hidden layers or neurons increase model complexity.
    - The network memorizes the training samples instead of learning general patterns.
3. **Insufficient Training Data**
    - A small dataset does not represent all possible student score patterns.
    - This reduces the model's ability to generalize.
4. **Poor Data Quality**
    - Missing values, outliers, or inconsistent data can negatively affect prediction accuracy.
    - Differences between training and testing data distributions may also increase test error.
5. **Lack of Regularization**
    - Without techniques such as dropout or L2 regularization, the model is more likely to overfit.

Corrective Measures

1. **Increase the Training Dataset**

- Collect more student records or apply suitable data augmentation techniques where applicable.

2. **Use Regularization**
   - Apply L1/L2 regularization and dropout layers to reduce overfitting.
3. **Simplify the Model**
   - Reduce the number of hidden layers or neurons to avoid unnecessary complexity.
4. **Apply Early Stopping**
   - Stop training when the validation loss starts increasing, preventing the model from memorizing the training data.
5. **Perform Proper Data Preprocessing**
   - Handle missing values, remove outliers, and normalize or standardize input features.
6. **Use Cross-Validation**
   - Evaluate the model on multiple data splits to ensure better generalization.

Conclusion

The significant gap between the training MSE (**0.85**) and testing MSE (**5.20**) suggests that the neural network suffers from **overfitting**. By improving data quality, simplifying the model, applying regularization, using early stopping, and increasing the amount of training data, the model can achieve better performance on unseen data and reduce the testing error.

Code:

```python
# Error Analysis of Neural Network

training_mse = 0.85
testing_mse = 5.20

print("Training MSE:", training_mse)
print("Testing MSE:", testing_mse)

difference = testing_mse - training_mse
print("Difference:", difference)

if testing_mse > training_mse * 2:
    print("\nAnalysis:")
    print("The model is overfitting.")
    print("It performs well on training data but poorly on test data.")

    print("\nPossible Reasons:")
    print("- Overfitting")
    print("- Complex neural network")
    print("- Small training dataset")
    print("- Poor data quality")
    print("- Lack of regularization")

    print("\nCorrective Measures:")
    print("- Increase training data")
    print("- Apply Dropout or L2 Regularization")
    print("- Simplify the neural network")
    print("- Use Early Stopping")
    print("- Normalize and clean the dataset")
    print("- Perform Cross Validation")
else:
    print("\nThe model generalizes well.")
```

Output:

```
Training MSE: 0.85
Testing MSE: 5.2
Difference: 4.3500000000000005

Analysis:
The model is overfitting.
It performs well on training data but poorly on test data.

Possible Reasons:
- Overfitting
- Complex neural network
- Small training dataset
- Poor data quality
- Lack of regularization

Corrective Measures:
- Increase training data
- Apply Dropout or L2 Regularization
- Simplify the neural network
- Use Early Stopping
- Normalize and clean the dataset
- Perform Cross Validation
```

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided